

[IS216] Extra Exercises – Week 1 - JavaScript Syntax and Operations

Objectives

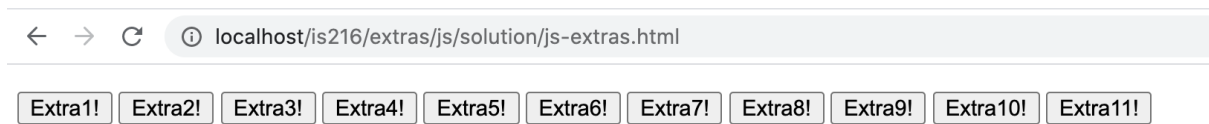
- To get familiar with JavaScript syntaxes
- To practice on applying JavaScript operations to solve programming problems

Instructions

- Questions with no asterisk mark are easy.
- Questions marked with * are slightly challenging.
- Questions marked with ** are challenging.
- Questions marked with *** are very challenging.

NOTE: If you spot any mistakes/errors in the questions, please contact your instructors by email and state the issues. We will try to address it as soon as possible.

All the following exercises can be done in `js-extras.html` in resource folder



Question 1 (*)

Write a function that computes and prints out the area and circumference of a circle of a given radius.

The radius could be set to an int or float value. You can set the value of π to 3.14

Formulas:

- Area of circle = $\pi \times \text{radius}^2$
- Circumference of circle = $\pi \times 2 \times \text{radius}$

Hint: explore `Math.PI()`

See examples below:

localhost says

Provide a radius!

Cancel

OK

← → ↻ ⓘ localhost/is216/extras/js/solution/js-extras.html

Area: 50.27

Circumference: 25.13

Question 2 ()**

Write a function that generates a random password of 8 characters.

For example, password -> cxqaptgh

Hint: explore `Math.random()`, `Math.floor()`

← → ↻ ⓘ localhost/is216/extras/js/solution/js-extras.html

Generated Password: lsbgmgwf

Question 3 (*)

In number theory, a perfect number is a positive integer that is equal to the sum of all its factors excluding itself.

For example, 6 is a perfect number because the sum of its factors i.e. $1 + 2 + 3 = 6$

Write a function that takes in a positive integer. The function should then check if the number entered is a perfect number or not and print the result. You can assume that only positive integers are entered.

Some examples of perfect numbers are 6, 28, 496

The user enters number '4' localhost says Please provide a number! 4 Cancel OK	Number '4' is not a perfect number: Not Perfect Number!
The user enters number '6' localhost says Please provide a number! 6 Cancel OK	Number '6' is a perfect number: Perfect Number!

Question 4 ()**

Write a function that converts a given integer into its equivalent binary number.
For example, 10 -> 1010, 12 -> 1100

Your function must validate that the user enters a valid integer. That is, ask the user to enter an integer again if she/he enters a non-integer value, e.g. 'abc'

Hint: explore `Number()`, `Number.isInteger()`, `toString()`

User enters a non-integer value 'abc' localhost says Please provide an integer! abc Cancel OK	The program prompts the user to enter a valid integer: localhost says Please provide a valid integer! Cancel OK
User enters a decimal value '2.4' localhost says Please provide an integer! 2.4 Cancel OK	The program prompts the user to enter a valid integer: localhost says Please provide a valid integer! Cancel OK
User enters an integer '68' localhost says Please provide an integer! 68 Cancel OK	Its equivalent binary number: 1000100

Question 5 ()**

Write a function that requests the user's name so that it displays back with the surname in capital letters and the rest of the name with first letter in uppercase.

Assume the following:

- a) Two adjacent words of the name is separated by 1 space
- b) If the name has 1 word, it is the surname
- c) If the name has 2 words, the second word is the surname
- d) If the name has more than 2 words, the first word is always the surname

For example, tan -> TAN, tan wee kiatt -> TAN Wee Kiat, mary lim -> LIM Mary

Hint: use `toUpperCase()`, `charAt()`

The name that has 1 word localhost says Please provide your name! tan Cancel OK	The result TAN
The name that has 2 words localhost says Please provide your name! mary lim Cancel OK	The result LIM Mary
The name that has 3 words localhost says Please provide your name! tan wee kiatt Cancel OK	The result TAN Wee Kiat

Question 6 ()**

Write a function that asks the user to enter a string, say `txt`, and two characters, say `st` and `en`.

The function then searches for a substring in `txt` such that the substring begins with the character `st` and ends with the character `en`.

If there is no such substring, display the message "No such substring".

Assume that the search is case-sensitive and only search for the first occurrence of the substring

Hint: explore `indexOf`, `search()`, `includes()`

For example,

```
txt = Fibonacci series
```

```
st = F
```

```
en = b
```

==> Substring [Fib] is found

```
txt = Fibonacci series
```

```
st = b
```

```
en = y
```

==> No such substring

```
txt = Fibonacci series
```

```
st = F
```

```
en = F
```

==> Substring [F] is found

Question 7 ()**

Write a function that asks the user to enter two strings, say `str1` and `str2`. The function displays "Bingo!" if every character in `str1` also appears in `str2`. Otherwise, it displays "Nope :(")

Hint: explore `includes()`

For example,

```
str1 = daily
str2 = Holiday
=> Bingo!
```

```
str1 = lily
str2 = Holiday
=> Bingo!
```

```
str1 = pokemon
str2 = pogo
=> Nope :(
```

Question 8 ()**

Write a function that requests two integers, say `min` and `max`. The function then displays all the perfect squares between `min` and `max`, inclusive.

For example, given `min=10` and `max=110`, it displays the perfect squares – 16 25 36 49 64 81 100.

Your function must validate that the user provides only numbers (decimal numbers, e.g., 10.2, are accepted).

Hint: explore `Math.sqrt()`, `Math.floor()`, `Math.ceil()`, `Math.pow()`

The user enters an invalid number localhost says Please enter a min number! abc Cancel OK localhost says Please enter a max number! 100 Cancel OK	The result localhost says You did not enter valid number! OK
The user enters a min number '9' and a max number '109' localhost says Please enter a min number! 9 Cancel OK localhost says Please enter a max number! 109 Cancel OK	The result 9 16 25 36 49 64 81 100

The user enters a min decimal number
'10.2' and a max number '100'

localhost says

Please enter a min number!

Cancel

OK

localhost says

Please enter a max number!

Cancel

OK

The result

16 25 36 49 64 81 100

Question 9 (*)**

Write a function that gets a set of integer inputs from the user. Assume that the user enters each integer, separated by a space. For example, 6 12 4 10

Your function validates that the user enters valid integers.

It should then display the minimum, maximum and median of all numbers entered.

Note: The median is the middle of the list of numbers.

For example, the median of numbers 12, 4, 5 is 5. In case of odd amount of numbers, the median is the exact middle number of numbers when arranged sorted. In case of even amount of numbers, we would get a pair of middle numbers. The median is half way between them.

For example,

median of numbers 6, 12, 4, 10 is 8 $(6 + 10) / 2$

because when placed in order the middle numbers would be 6 and 10.

Hint: explore `split()`, `sort()`

The user enters 4 valid integers: localhost says Please enter a set of integer numbers, each number separated by a space! 6 12 4 10 Cancel OK	The result: Numbers: 4,6,10,12 Min: 4 Max: 12 Median: 8
The user enters 7 valid integers: localhost says Please enter a set of integer numbers, each number separated by a space! 9 11 71 2 44 90 13 Cancel OK	The result: Numbers: 2,9,11,13,44,71,90 Min: 2 Max: 90 Median: 13
The user enters an invalid integer:	The result:

localhost says Please enter a set of integer numbers, each number separated by a space! 1 2 3 a Cancel OK	localhost says You entered an invalid integer! OK
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Question 10 (***)

Write a function that merges two arrays of objects, representing persons. Each person object has name and age properties. Assume that names are unique.

Sample arrays of objects are given in `js-extras.html`

```
function showExtra10() {  
    var persons1 = [{name:"John",age:29}, {name:"Jesper",age:21}, {name:"Jack", age:40}];  
    var persons2 = [{name:"Mary",age:23}, {name:"Jennifer",age:32}];  
}
```

The function merges the two arrays into a single array in which the persons are ordered by their age.

Also create your own arrays of objects in the function to test your code.

Hint: explore `concat()`

Example:

Name: Jesper; Age: 21
Name: Mary; Age: 23
Name: John; Age: 29
Name: Jennifer; Age: 32
Name: Jack; Age: 40

Question 11 (**)

Write a function that computes the total amount to be paid, based on an array of purchase items. A purchase item is an object, having **name** and **price** properties. For example, name="pencil", price=1.25

Create your own arrays of objects in the function.